Tokens

The elements required to develop program is called as Tokens
Tokens are 5 types, They are

1. Literals
2. Constants
3. Operators
4. Keywords
5. Identifiers

1. Literals : The values in our program is called as literals

--> Literals are 4 types, they are

1.Integer Literals : Any number without decimal part is called as integer

Ex: age = 25 empid = 101 tokenno = 5

2.Floating Literals: Any number with decimal part is called as Floating

Ex: height = 4.8 pcost = 48.23 weight = 62.32

3.String Literals : Anything which is enclosed with in single or double quotes

Ex: name='AVD' or name="AVD"
qual = 'MCA,M.Tech' doj = '02-Nov' a ='5'

4.Boolean Literals: True, False are called as Boolean Literals

--> At the time of conditions we need to use boolean literals

Check Person is Eligible for vote or not ---> age >=18

2. Constants : The values which are fixed is called as constants

Ex: PI = 3.14

noofmonthsperyear = 12

3. Operators : Anything which performs an operation is called as Operator

Ex: 3 + 5 ---> '+' is Operator , 3,5 are Operands
a < b ---> '<' is Operator, a,b are Operands
p $ q ---> '$' is Symbol

-->In Python we have different types of Operators, they are

1. Arithmetic Operators : +, -, *, /, //, %, **
2. Relational Operators : <, <=, >, >=, ==, !=
3. Logical Operators : and, or, not
4. Assignment Operator : =
5. ShortHand Assignment Operators : +=, -=, *=, /=, //=, %=, **=
6. Memebership Operators : in, not in

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Arithmetic Operators

--> If you want to perform any calculations then use Arithmetic Operators

Operators	Meaning	Ex
+	Addition	7 + 2 ---> 9
-	Subtraction	7 - 2 ---> 5
*	Multiplication	7 * 2 ---> 14
/	Division	7 / 2 ---> 3.5
//	Floor Division	7 // 2 ---> 3
%	Modulo Division	7 % 2 ---> 1
**	Exponent/Power	7 ** 2 ----> 49

2) 7 (3
6

1

Keywords

--> Keywords are Pre-Defined words

--> It is also called as Reserved Words, Every keyword will perform some operation

```
import keyword
keyword.kwlist
```

And ⇨ Not a keyword

```
['False',     'None',     'True',     'and',     'as',
 'assert',    'async',    'await',    'break',   'class',
 'continue',  'def',      'del',      'elif',    'else',
 'except',    'finally',  'for',      'from',    'global',
 'if',        'import',   'in',       'is',      'lambda',
 'nonlocal',  'not',      'or',       'pass',    'raise',
 'return',    'try',      'while',    'with',    'yield']
```

--> False, None, True are values they never perform any operations

Identifiers

--> Identifiers are User-Defined words

--> The names given to Variables, functions, methods, classes, objects etc are called as Identifiers

Identifiers

```
class  BookTicket:              class  CancelTicket:
    ......................          ......................
    ......................          ......................
```

--> An Identifier given to class is called as classname

--> An Identifier given to function is called as function name

--> An Identifier given to object is called as object name

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--> For Identifiers we can give any name, but we need to follow 4 rules

Rule1: No Space

Book Ticket × Invalid, Because it contain space

Rule2: No Special Symbols except underscore(_)

Book@Ticket × Invalid, Because it contain '@'
Book_Ticket
_Book_Ticket_

Rule3: No Keywords, That means we can not use keyword names as identifiers

and × Invalid, Because 'and' is a keyword

And ✓ Valid, Because And is not a keyword

Rule4: First character should be either alphabet or Underscore(_)

2grade × Invalid, Because first character is number

_2grade
grade2 } Valid

Identify Invalid Identifier Names:

i)customer name	ii) salary	iii) sub1_marks
iv) _eid	v) 7bc	vi) true
vii) loanAmount	viii)Else	ix) course
x) no.of.seats	xi) leaves	xii) if
xiii) discount_Amount	xiv)gstAmt	xv) taxAmt
xvi) calTotalMarks	xvii)Employee	xviii)Loan
xix) interest-Amount	xx)customer2	xxi) pid

Variables:

- ➔ Variable is a named memory location
- ➔ Variables are used to store values
- ➔ We can store values by using Assignment Operator (=)
- ➔ Assignment Operator is also called as Right to Left operator

Syntax: $\underbrace{\text{Variablename}}_{\text{Left}} = \underbrace{\text{Value}}_{\text{Right}}$

Ex: age = 22

age → Variable name

$\boxed{22}$ → Value

pcost = 256.26

name = ‘AVD’

Value

Syntax2: Variablename = $\boxed{\text{Value / Operand / Expression}}$

Ex: a = 5

b = a

5

a	b	c
5	5	10

c = p --> Error, Because we don't have any variable with 'p'

c = a + b